

Learning Objective 2.4: I can describe how reflectors, Yagi-Uda antennas, and arrays achieve high gain, and select an appropriate high-gain antenna for a given application.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Sizing a C-band earth station.** A ground station uses a **3.0 m diameter** prime-focus parabolic reflector at **6.0 GHz**. Take the aperture efficiency to be $\eta_{\text{ap}} = 0.65$ and $c = 3 \times 10^8$ m/s.

- (a) In one or two sentences, explain why the statement “gain is really a statement about area” is more useful to a designer than “gain is how much the antenna amplifies.”

- (b) Find the wavelength, the dish diameter in wavelengths, and the gain in dBi.

$$G =$$

- (c) Estimate the half-power beamwidth, and state in one sentence what it implies about pointing.

$$\theta_{HP} =$$

- (d) Management wants **3 dB more gain** at the same frequency. What dish diameter does that take, and name one specific cost of the change.

2. **Where the efficiency went.** Same 3.0 m dish. The manufacturer quotes the following loss factors: spillover 0.88, illumination taper 0.86, blockage 0.96, and a lumped “everything else” factor of 0.98. The reflector surface is held to **0.5 mm RMS**. Ruze’s formula gives the surface penalty as loss (dB) = $685.8 (\sigma/\lambda)^2$.

- (a) Spillover and illumination taper trade directly against each other. Explain the trade in one or two sentences, and state the rule of thumb for where the optimum sits.

- (b) Compute the aperture efficiency from the four quoted factors, before any surface error. Call it $\varepsilon_{ap,1}$.

$$\varepsilon_{ap,1} =$$

- (c) Add the Ruze surface term at 6.0 GHz and report the total aperture efficiency.

$$\eta_{ap} =$$

- (d) Someone proposes reusing this *same physical dish* at **30 GHz**. Recompute the surface term and the total aperture efficiency, then say in one or two sentences what the result means for the proposal.

3. **Reading a Yagi-Uda.** A Yagi-Uda has one driven element, one reflector behind it, and a row of directors in front along a boom.

(a) The reflector is cut slightly *longer* than resonance and the directors slightly *shorter*. Explain what that detuning does electrically, and how it produces an endfire beam.

(b) A second reflector buys almost nothing, but a seventh director still helps. Why?

(c) A 6-element Yagi on a 1.0λ boom gives 10 dBi. Using the rule of thumb of +3 dB per doubling of boom length, estimate the boom needed for 16 dBi at 435 MHz. Then say whether you believe the answer.

$L =$

(d) Yagis are narrowband — a few percent. Explain why, and name one application where that does not matter.

4. **Choosing an antenna.** You need a **25 dBi** antenna at each end of a 2 km rooftop-to-rooftop link at **5.8 GHz**. The mounts are fixed (no steering), the site is windy, and the budget is small.

- (a) Find the required effective aperture, then the diameter of a parabolic dish that delivers it with $\eta_{\text{ap}} = 0.60$.

$D =$

- (b) A vendor offers a flat patch array instead, with $\eta_{\text{ap}} = 0.75$ and elements on a $\lambda/2$ grid. How many elements, and how big is the panel?

$N =$

- (c) Pick one and defend it in two or three sentences, using the numbers you just computed.

- (d) The dish beam is 9.6° wide. Using the beam-shape approximation $L(\theta) \approx 3(2\theta/\theta_{HP})^2$ dB, how far off bore-sight can the mount drift before you lose 0.5 dB?

$\theta =$

5. **Does the link close?** Continue the 5.8 GHz, 2 km link from the previous question. The transmitter delivers **20 dBm** and the receiver sensitivity is **−80 dBm**. Use Friis, $P_r = P_t G_t G_r (\lambda/4\pi R)^2$.

(a) Compute the free-space path loss in dB.

$$L_{fs} =$$

(b) Find the received power with the 25 dBi dishes at both ends, and the link margin.

$$P_r =$$

(c) Now replace both dishes with 8 dBi patches. Recompute P_r and the margin.

$$P_r =$$

(d) Which link would you actually deploy, and why? Name two specific things that eat margin on a real rooftop link.

Documentation: _____